

Practice Questions

Test your comprehension by answering the following questions:

1. What is a benefit of the divide-and-conquer summing algorithm in this chapter compared to the head-tail summing algorithm in Chapter 3?
2. If a binary search of 50 books on a shelf takes six steps, how many steps would it take to search twice as many books?
3. Can a binary search algorithm search an unsorted array?
4. Is partitioning the same thing as sorting?
5. What happens in quicksort's partition step?
6. What is the pivot value in quicksort?
7. What is the base case of quicksort?
8. How many recursive calls does the `quicksort()` function have?
9. How is the array `[0, 3, 1, 2, 5, 4, 7, 6]` not properly partitioned with a pivot value of 4?
10. What is the base case of merge sort?
11. How many recursive calls does the `mergeSort()` function have?
12. What is the resultant array when the merge sort algorithm sorts the arrays `[12, 37, 38, 41, 99]` and `[2, 4, 14, 42]`?
13. What is a lookup table?
14. In the Karatsuba algorithm that multiplies integers x and y , what do the a , b , c , and d variables store?
15. Answer the three questions about recursive solutions for each of the recursive algorithms presented in this chapter:
 - a. What is the base case?
 - b. What argument is passed to the recursive function call?
 - c. How does this argument become closer to the base case?